ECO 344 - Applied Econometrics I
Chapter 7

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Outline

1. Introduction
2. Dummy Variable as an Independent Variable
3. Using Multiple Dummy Variables as Independent Variables
4. Interaction Terms for Dummy Variables
5. Binary Variable as a Dependent Variable
6. Policy Analysis and Program Evaluation
7. Discrete Dependent Variable
8. Summary

Introduction

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- Be careful while including such variables in the analysis and while interpreting them.

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- We can include the dummy variable as an independent variable in the regression. Let us see what this does to our model.

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- In the example above, δ_f is the difference in the average wage of a female and a male with the same education.
- If $\delta_f < 0$, females earn less than comparable males.
- Under what condition can we say that there is no gap between male and female wages? ($\delta_f = 0$)

Dummy Variable: Intercept Change

- Can we call this an intercept change?

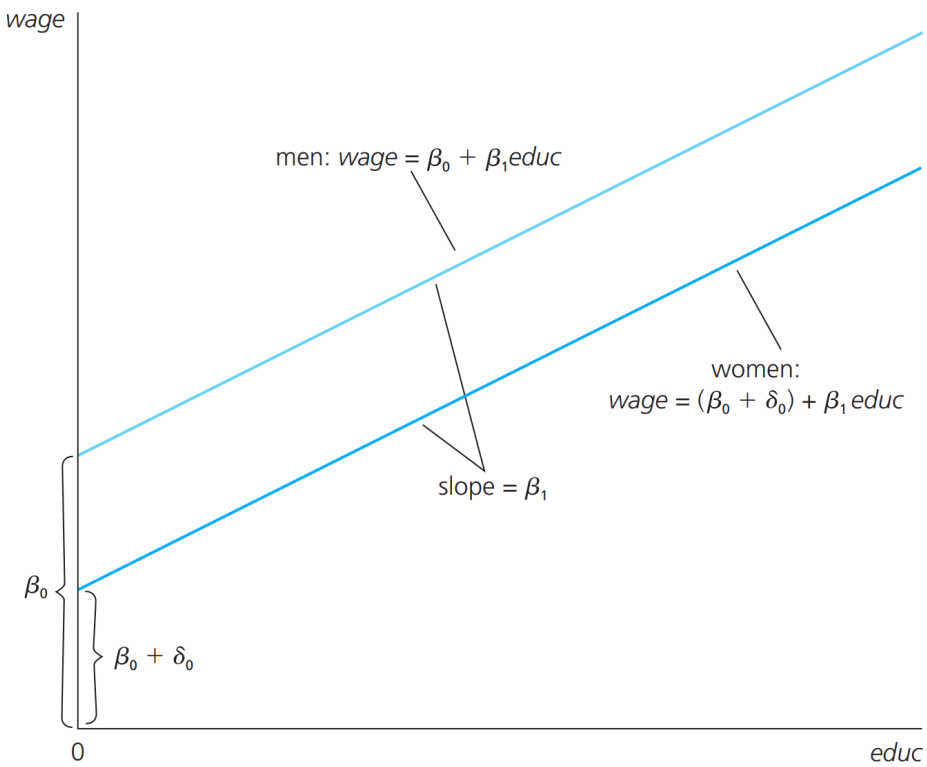
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FIGURE 7.1 Graph of $wage = \beta_0 + \delta_0\, female + \beta_1\, educ$ for $\delta_0 < 0$.



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- Generally, the coefficient of a dummy variable tells us the difference in y as compared to the base group (the group where the dummy variable is zero).
- The choice of base group does not change the fit of the model. It only changes how the coefficients are read.

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- So the dummy gives us a conditional gap. A causal claim about discrimination needs more than adding a dummy to the regression.

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- This approximation is good only when δ_f is small. When the coefficient is large, the log approximation is poor.
- The exact percentage difference between the group coded 1 and the base group is $100 \times (e^{\delta_f} - 1)\%$.

Example 1: Hourly Wage Equation (Example 7.1)

- Estimating the model we have been discussing, with experience and tenure added as controls, gets us the following sample regression function.

$$\widehat{wage} = -1.57 - 1.81 female + 0.572 education + 0.025 exper + 0.141 tenure$$

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- Is this an indication of discrimination? Remember the previous slide.

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- Is this a causal effect of the computer? What might be missing from the model?

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- Let us consider the following regression for the wage gap between females and males.

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- Approximately, a female earns 29.7 percent less than a male with the same education, experience and tenure.
- The exact difference is $100(e^{-0.297} - 1) = -25.7$. So a female earns about 25.7 percent less than a comparable male.

Using Multiple Dummy Variables as Independent Variables

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- What is the interpretation of β_1 in this case?
- Does the interpretation feel strange? It assumes that every step up the code has the same effect on wage.

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- All the discussion regarding a log dependent variable applies here as well.
- To compare two included groups, for example master's and undergraduate, look at $\beta_5 - \beta_4$. A t-test on this difference needs the standard error of the difference, or we can simply change the base group.

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- Here the base group is Poor, and β_3 is the difference in the interest rate between an Excellent and a Poor rating.

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- In this example, we drop the dummy of "others" so that it becomes the base group. The coefficient of each party is then read against all the small parties.

Interaction Terms for Dummy Variables

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- What do β_1 and β_2 represent?
- This model forces the effect of marriage to be the same for males and females. What if we want to see the effect of being a married female?

Interaction Between Dummy Variables

- To capture the effect of being a married female, let us consider the following model:

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- Now the base group is clear: unmarried males. Each of the four groups has its own average wage. How? Write the possibilities on paper.

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  - So β_3 is the extra/lower wage of a married female beyond the two individual effects.

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- It is not necessary that the slope is different for males and females. How would that show up in our estimation?

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  - β_0 is the average wage of a male with zero education.

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$$wage = \beta_0 + \beta_1 education + \mu$$

  - Therefore, β_1 is the effect of education on the wages of males.
  - β_0 is the average wage of a male with zero education.
- When female $= 1$

$$wage = (\beta_0 + \delta_0) + (\beta_1 + \delta_1) education + \mu$$

Allowing for Different Slope

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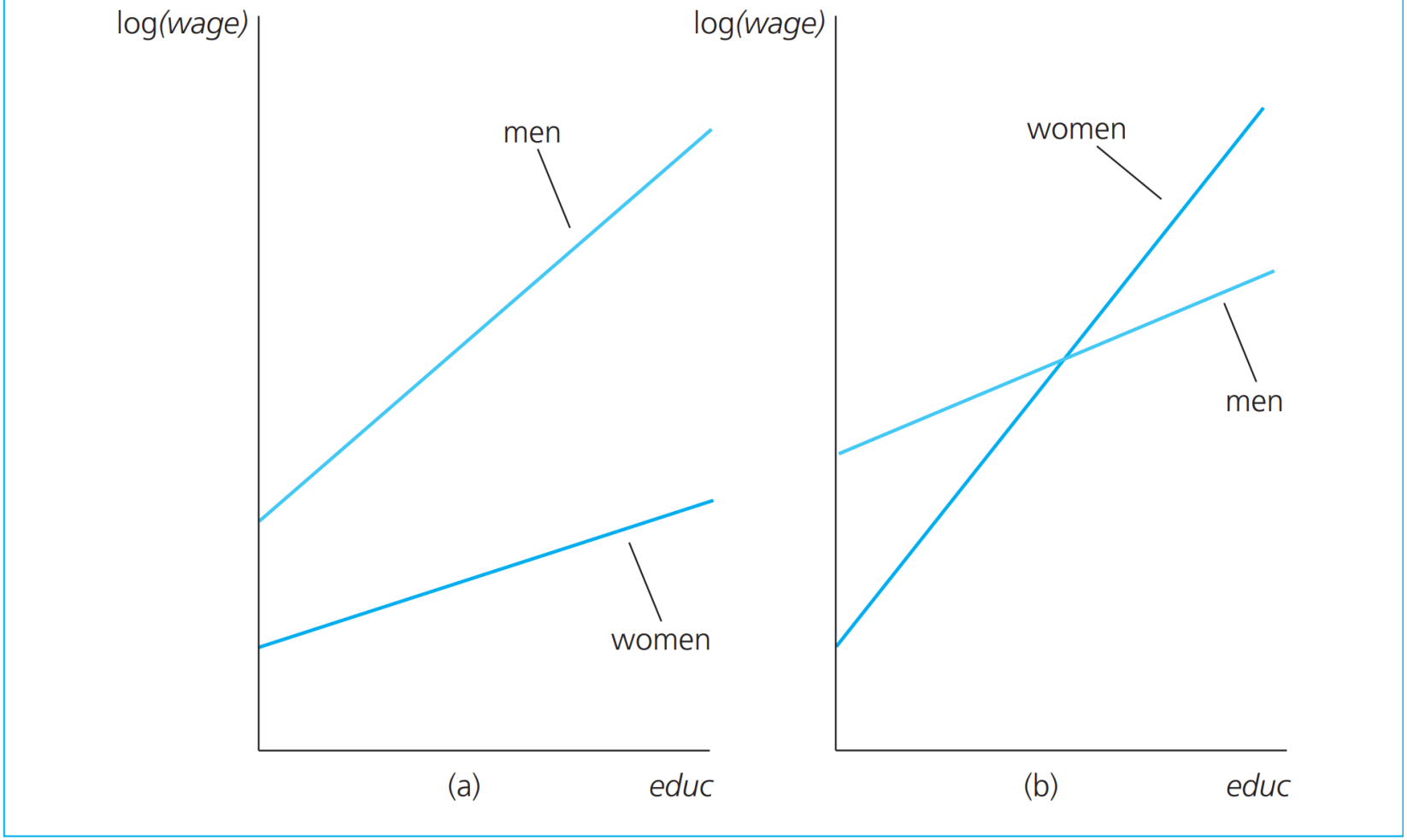
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FIGURE 7.2 Graphs of equation (7.16): (a) $\delta_0 < 0, \delta_1 < 0$; (b) $\delta_0 < 0, \delta_1 > 0$.



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- The interaction coefficient is small and statistically insignificant, so we cannot reject that the returns are equal.
- The female coefficient -0.227 is the gap at $educ = 0$. Nobody in the sample has zero education, so this number has no practical meaning on its own.
- A better question is the gap at a meaningful education level, for example 12 years: $-0.227 - 0.0056 \times 12 \approx -0.29$.

Testing for Difference in Groups

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$$F = \frac{\frac{SSR_r - SSR_{ur}}{q}}{\frac{SSR_{ur}}{df_{ur}}} = \frac{\frac{SSR_r - SSR_{ur}}{k+1}}{\frac{SSR_{ur}}{n-2(k+1)}}$$

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4. If $F > F_{k+1,\, n-2k-2}$ (critical value), we reject H_0 and say that both groups have different regression functions. Otherwise, we conclude there is no evidence of a difference between the regression functions of males and females.

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- Caution: this form of the test assumes the error variance is the same in both groups. If it is not, use the dummy and interaction version with heteroskedasticity robust standard errors.

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- Similarly, each additional child under six changes the probability of being in the labor force by β_2, keeping all other factors constant.

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- The LPM is easy to use and interpret, but it has some problems.

Example: Women's Labor Force Participation

- A linear probability model for women's labor force participation (753 women) gives:

$$\widehat{inlf} = 0.586 - 0.0034\, nwifeinc + 0.038\, educ + 0.039\, exper - 0.0006\, exper^2 - 0.016\, age - 0.262\, kidslt6 + 0.013\, kidsge6$$

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- Each additional child under six lowers the probability of working by 0.262, that is 26.2 percentage points.
- What happens to the predicted probability for a woman with a very low income, a lot of experience and no young children? Could it exceed 1?

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- Logit and probit models fix the first two problems. We will not cover them here.

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- Only if participation is unrelated to the unobserved factors in μ once the controls are included.

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Policy Analysis and Program Evaluation

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- This is why randomized assignment is the best case, and why we add controls when assignment is not random.

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  - The same discussion applies to the predicted values. A prediction of 2.7 is an expected number of children, not a count we will ever observe for one person.

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- With a binary dependent variable, OLS gives the Linear Probability Model. Read coefficients as changes in probability and use robust standard errors.
- A program dummy gives the causal effect only if participation is unrelated to the unobserved factors.